Lesson 13A Newton's Law of Gravitation

Introduction to Gravitation

Gravity is a fundamental force of nature—one that shapes the universe on every scale. It:

- Holds us on Earth, and Earth in orbit around the Sun
- Keeps the Sun bound to the stars in our galaxy
- Binds galaxies together in clusters and superclusters
- Drives the collapse of massive stars into black holes
- Attempts to slow the expansion of the universe

Despite being the weakest of the four fundamental forces, gravity's reach is immense and universal.

Newton's Law of Universal Gravitation

Sir Isaac Newton proposed that **any two objects with mass attract each other** with a force that depends on both their masses and the distance between them. This force is always **attractive**, never repulsive.

The **magnitude** of the gravitational force between two point masses m_1 and m_2 separated by a distance r is given by:

$$F = G\frac{m_1 m_2}{r^2}$$

- F is the gravitational force
- G is the **gravitational constant**, approximately

$$G = 6.674 \times 10^{-11}\,\mathrm{N}\cdot\mathrm{m}^2/\mathrm{kg}^2$$

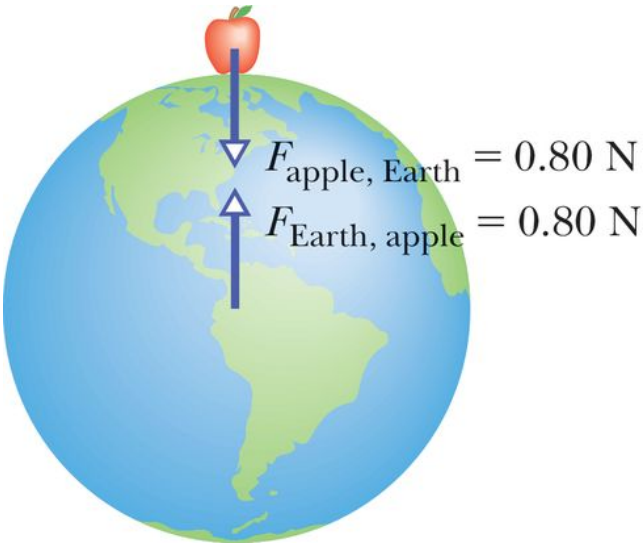
This can be expressed in **vector form** to include direction:

$$\vec{F}_{12} = -G\frac{m_1 m_2}{r^2}\,\hat{r}_{12}$$

Here, $\hat{r}_{12}$ is the **unit vector** pointing from mass 1 to mass 2, and the negative sign indicates the **force is attractive**.

Mass, Attraction, and Acceleration

Gravitational attraction depends on how much mass an object contains. Earth's large mass creates a strong gravitational pull. However, this force forms **Newton's third-law pairs**:



- The Earth pulls on an apple with a force of about **0.8 N**
- The apple pulls on the Earth with the **same** force in the opposite direction

The difference in mass leads to different **accelerations**:

- Apple: $a = \frac{F}{m_{\text{apple}}} \approx 9.8\,\text{m/s}^2$
- Earth: $a = \frac{F}{m_{\text{Earth}}} \ll 1\,\text{m/s}^2$

The Shell Theorem

Newton's **shell theorem** explains how gravity behaves for spherically symmetric mass distributions (like planets):

- A spherical shell exerts no net gravitational force on an object inside it

- Outside the shell (or solid sphere), the gravitational force behaves as if all the mass were at the center

Thus, the Earth's gravity can be modeled as if all its mass were concentrated at a point at the center of the planet.

Superposition of Gravitational Forces

Gravity obeys the **principle of superposition**: the total gravitational force on an object is the **vector sum** of the individual forces from all other masses.

For multiple particles:

$$\vec{F}_{\text{net}} = \sum_i \vec{F}_i$$

For a continuous mass distribution, this becomes an integral:

$$\vec{F} = G \int \frac{dm \cdot m}{r^2} \, \hat{r}$$

In practice, if the object is a **uniform sphere**, the shell theorem allows us to treat its mass as if it were concentrated at the center, greatly simplifying calculations.